

# **Introduction to COM31006- Computer Vision**

## **Admin and Intro to CV**

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# Outline

- Module information
- Introduction to computer vision
  - Motivations
  - Various applications

# Teaching format

- Lecturers
  - Prof. Jungong Han, coordinator, will be teaching the first 7 weeks
  - Dr. Jefersson A. dos, will be teaching the last 4 weeks
  - Oakleigh Weekes, Xiaoqi Zhuang, Sijie Li (GTAs; PhD candidates)
- 11 x 2h lecture (11 weeks)
  - 1h (effectively 50 mins) x 2: Tuesday 9:00-11:00
- 11 x 2h lab/tutorial
  - 2 hours: Friday 16:00-18:00

# Content

| Week | Topic 1                                                     | Topic 2                                                     |
|------|-------------------------------------------------------------|-------------------------------------------------------------|
| 1    | Introduction about computer vision                          | Image formation                                             |
| 2    | Basic Image Arithmetic<br>and Point Operators               | Filtering and noise 1                                       |
| 3    | Filtering and noise 2                                       | Segmentation                                                |
| 4    | Edges                                                       | Edge detection: Canny detector                              |
| 5    | Feature extraction: Hough transform                         | Image transformation 1                                      |
| 6    | Image transformation 2                                      | Feature extraction and description 1                        |
| 7    | Feature extraction and description 2                        | Camera calibration and motion                               |
| 8    | Break                                                       | Break                                                       |
| 9    | Break                                                       | Break                                                       |
| 10   | Break                                                       | Break                                                       |
| 11   | Deep neural networks                                        | Network training: backpropagation                           |
| 12   | Convolutional Neural Networks for image<br>classification 1 | Convolutional Neural Networks for image<br>classification 2 |
| 13   | Advanced CNN Concepts and object<br>detection 1             | Advanced CNN Concepts and object detection 2                |
| 14   | Deep-Based Image Segmentation                               | FCNs and Encoder-Decoder Architectures                      |
| 15   | Exam briefing                                               | Exam briefing                                               |

# Assessment

- Assignment
  - Will be delivered in week 7 (18. April – 20. May)
  - Will have 5 weeks
  - 50% of the total mark
- Exam paper
  - 2h exam
  - 50% of the total mark

# 10 credits

- Corresponding to 100h of work
- 22 hrs lectures
- 22 hrs labs
- 2 hrs exam
- 54 hrs independent study (research driven module!)

# Prerequisites

- Required: decent programming skill (Python, C, C++)
- Not prerequisites, however, highly desirable to be skilled in:
  - Image processing
  - Linear algebra
  - Basic calculus
  - Basic probability
  - Basic statistics

# CV framework and programming language

- CV framework
  - OpenCV (> 3.4.2, preferably >= 4.0)
- Programming language
  - Python (deep learning especially)
  - C and simple C++



# Books

- Computer Vision: Algorithms and Applications, Richard Szeliski, Springer-Verlag Berlin, Heidelberg, 2010 Available online: <http://szeliski.org/Book/>
- Other books
  - Computer Vision: A Modern Approach, 2nd Edition, David A. Forsyth, and Jean Ponce
  - Image Processing, Analysis, and Machine Vision, 4th Edition, Milan Sonka, Vaclav Hlavac, and Roger Boyle
  - Vision : A Computational Investigation into the Human Representation and Processing of Visual Information, David Marr, Tomaso A. Poggio, and Shimon Ullman

# Other resources

- Discussion forum (BB)
- Ask us questions during the class room lectures/labs
- Email me and make an appointment for the discussion

# Module outcomes

- Express a consolidated and extended understanding and knowledge of CV techniques
- Explain the problems, techniques and difficulties associated with the different areas of CV
- Compare, critically evaluate and discuss competing methods

# Suggestions

- Attend the weekly lectures and labs
- Explore and play a lot with the software/frameworks used
- Contribute to the discussion forums

# How would you define (human) vision?

Merriam-Webster dictionary:

*the **special sense** by which the **qualities of an object** (such as color, luminosity, shape, and size) constituting its appearance are **perceived** through **a process** in which light rays entering the eye are **transformed** by the **retina** into **electrical signals** that are transmitted to the **brain** via the optic nerve*

# What is computer vision?

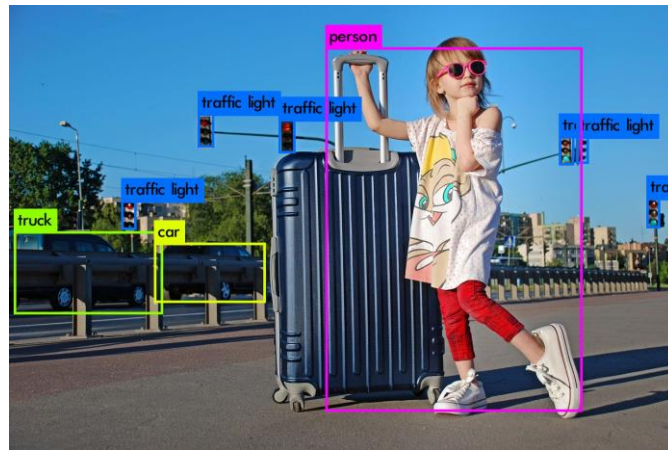
- Make computers understand images and videos for humans

What kind of scene?

How many objects?

Who is the girl?

....



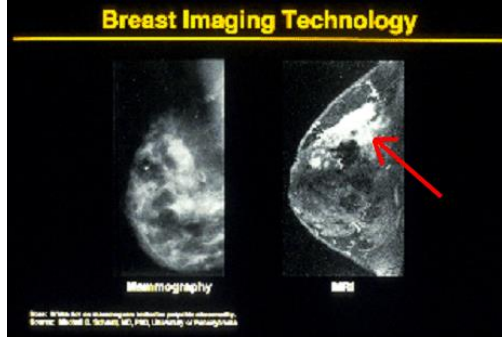
# Why computer vision?

- Images (and video) are everywhere
- Lots of exciting research problems
- One of the fastest growing technologies
- High demand from job market
- It's lots of fun

# How CV helps real world applications?



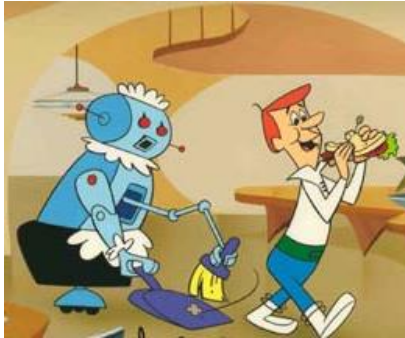
Safety



Health



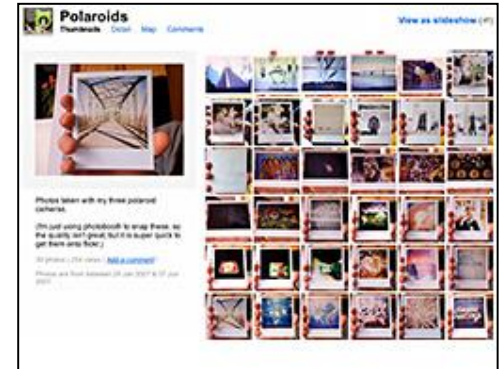
Security



Comfort



Fun



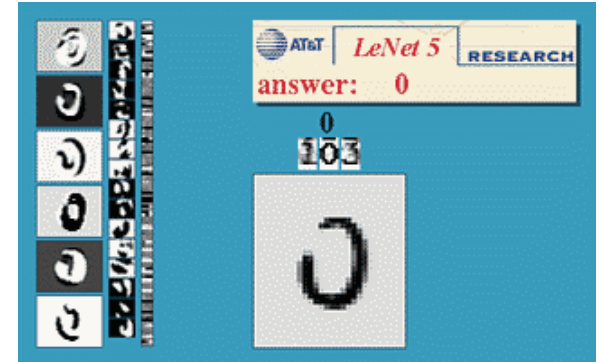
Access

[Slides from James Hays, Brown University]



# Optical character recognition (OCR)

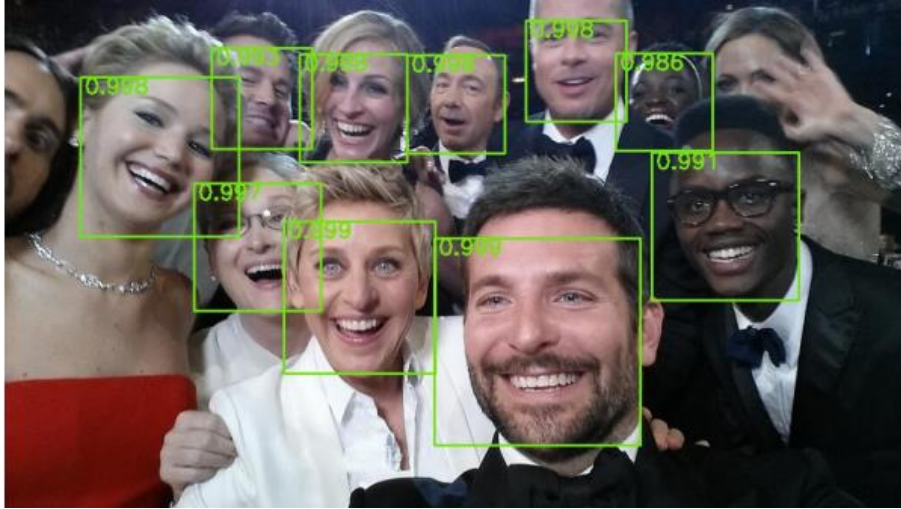
- Technology to convert scanned docs to text



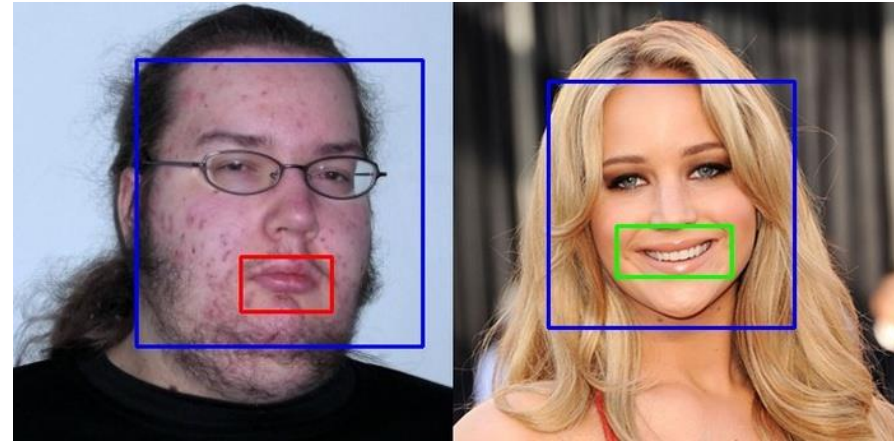
Digit recognition, AT&T labs  
<http://www.research.att.com/~yann/>

Source: <http://www.platerecognition.info>

# Face/smile detection



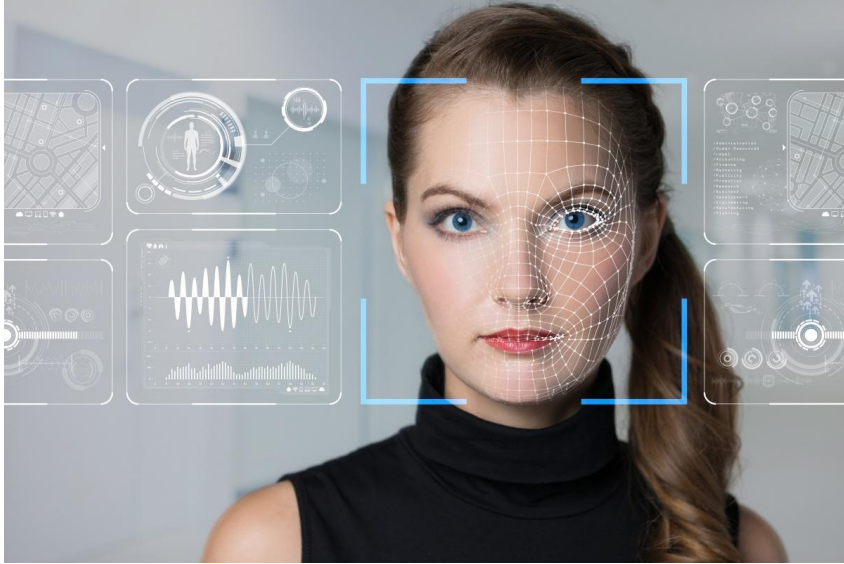
Source: <https://www.technologyreview.com>



Source: <https://facecrawler.weebly.com>

Check: <https://youtu.be/PcoJqV5E6Y>

# Facial/iris/retina recognition



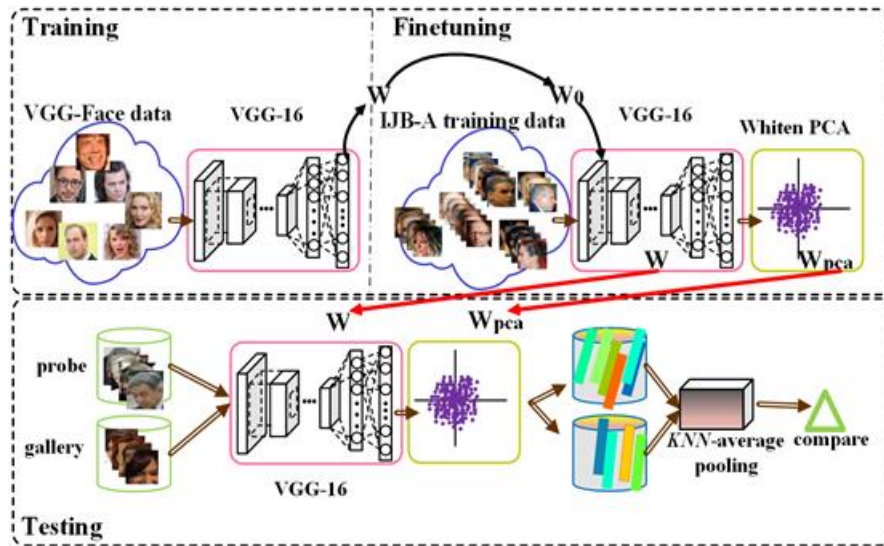
Source: <https://www.smartcitiesworld.net>



Source: <https://www.bayometric.com>

Check: <https://youtu.be/K4u4Dpl6NKk>

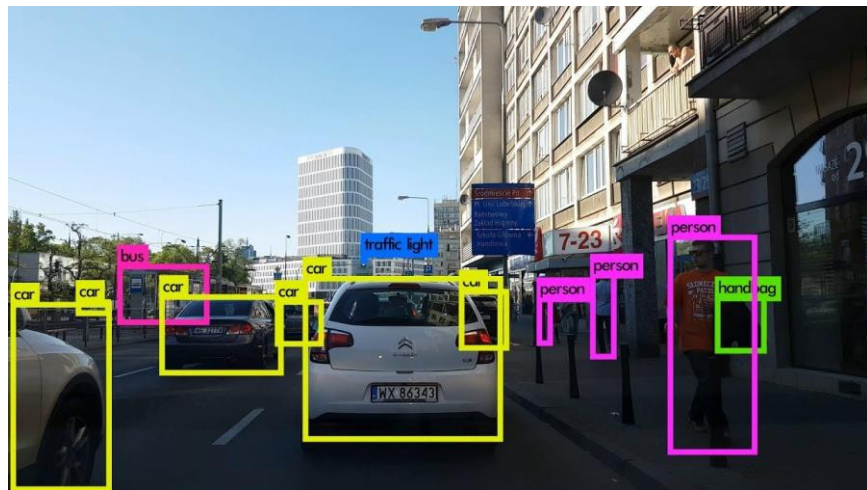
# Face recognition in wild



Zhao & Han & Shao, Unconstrained face recognition using a set-to-set distance measure on deep learned features, IEEE T-CSVT, 2017



# Object recognition/detection



Source: <https://blog.paperspace.com>



Source: <http://www.whippet.co.uk>

Check: <https://youtu.be/NrmMk1Myrxc>

# Smart cars

[▶ manufacturer products](#) [consumer products ◀](#)

## Our Vision. Your Safety.

rear looking camera

forward looking camera

side looking camera

**EyeQ** Vision on a Chip  
  
[read more](#)

**Vision Applications**  
Road, Vehicle, Pedestrian Protection and more  
  
[read more](#)

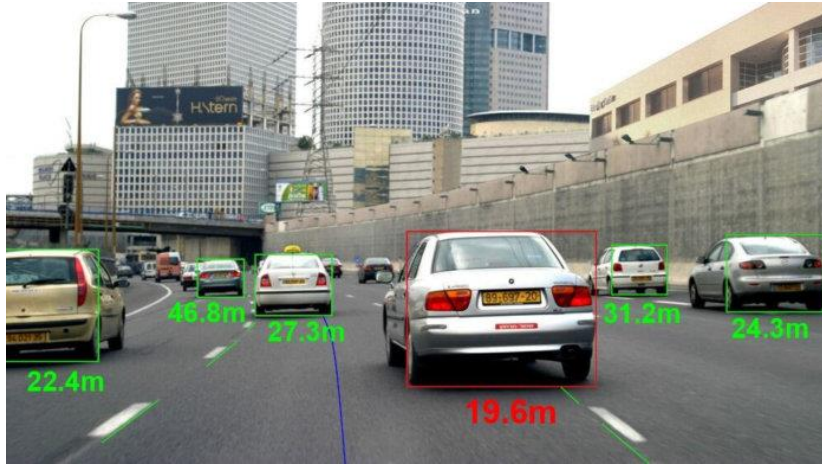
**AWS** Advance Warning System  
  
[read more](#)

**News**

- **Mobileye Advanced Technologies Power Volvo Cars World First Collision Warning With Auto Brake System**
- **Volvo: New Collision Warning with Auto Brake Helps Prevent Rear-end**
- [all news](#)

**Events**

- **Mobileye at Equip Auto, Paris, France**
- **Mobileye at SEMA, Las Vegas, NV**
- [read more](#)



Source: <https://www.computerworld.com>

# CV in sports

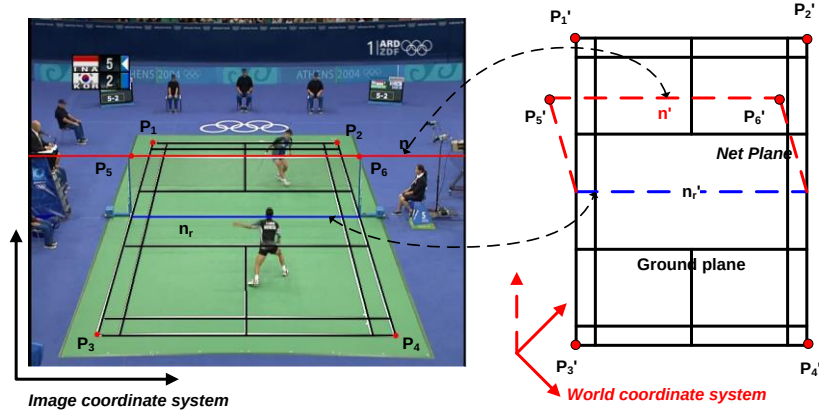


Source: <https://www.dailymail.co.uk>

Check: [https://youtu.be/dMLGp\\_NRgVQ](https://youtu.be/dMLGp_NRgVQ)



# 3D modeling for sports analysis



$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \approx \begin{pmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_{00} & r_{01} & r_{02} & t_x \\ r_{10} & r_{11} & r_{12} & t_y \\ r_{20} & r_{21} & r_{22} & t_z \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

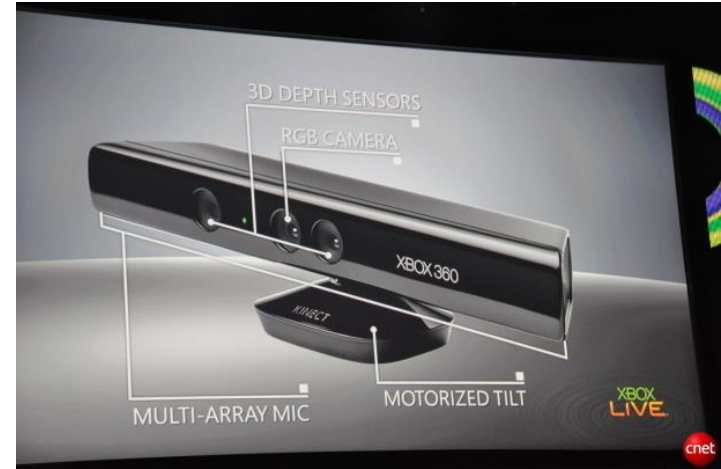
Projection Matrix

Camera Project    Camera Rotation  
and Placement    World  
Coordinate

Han et al. Broadcast Court-Net Sports Video Analysis Using Fast 3-D Camera Modeling, IEEE T-CSVT, 2008



# CV in games



Han & Shao & Xu & Shotton. Enhanced computer vision with microsoft kinect sensor: A review , IEEE T-Cybernetics, 2013  
[google citation > 2000](#)

# CV in robotics



Source: <https://www.technologyreview.com>



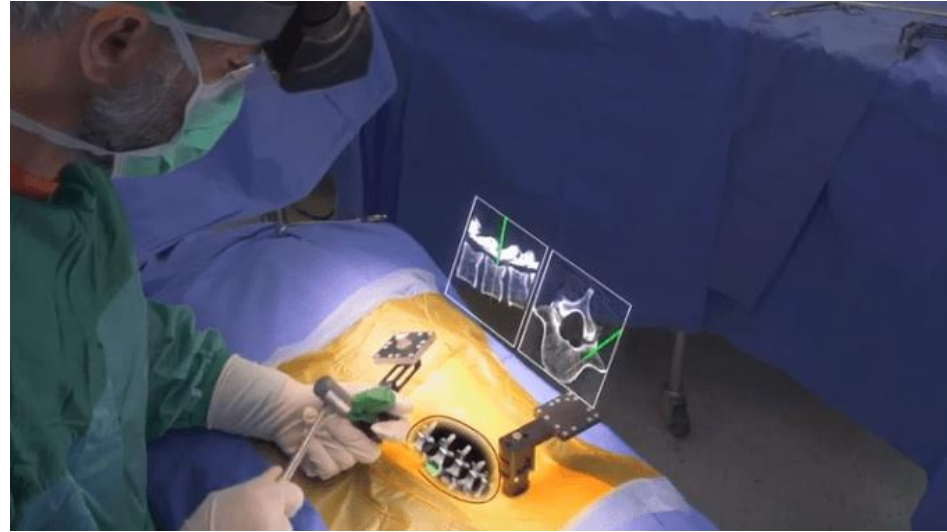
Source: Inventions World

Check: <https://youtu.be/kw37NUs6YKo>

# Medical imaging



Credits: MIT



Credits: Augmedics

# Computer vision vs Image processing

- Computer vision:
  - From image, return features/interpretation (i.e. meaning)
- Image processing:
  - From image, return image
  - Essential to Computer Vision
  - Many IP techniques classified in the “Early Vision” group

# Computer vision vs Computer graphics

- Computer vision:
  - Analysis (of images/videos)
  - From images to models
- Computer graphics:
  - Synthesis
  - From models to images

# Why implement CV algorithms

If libraries already have it implemented, why do we have to?

- So that you understand how it works
- So that when it doesn't work (using the library), you know why and how to fix it

# A note about the images used

In many presentations, images from the following resources are used:

- <https://pixabay.com>
- <https://wikipedia.org>
- [http://www.imageprocessingplace.com/root\\_files\\_V3/image\\_databases.htm](http://www.imageprocessingplace.com/root_files_V3/image_databases.htm) (highly recommended book for image processing)
- Any of the books from the reading list

# Summary

- Computer vision is not image processing
- Computer vision is not computer graphics
- Computer vision deals with extracting high-level understanding from images/videos
- You'll need to use all the remaining hours working for the module